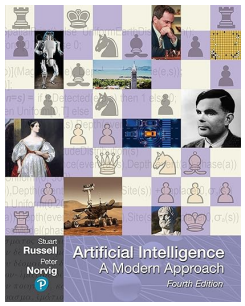
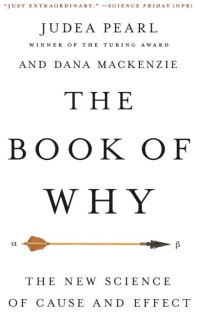
Lesson 08.4:

Instructor: Dr. GP Saggese, gsaggese@umd.edu

References:

- AIMA (Artificial Intelligence: a Modern Approach)
- Pearl et al., The Book of Why, 2017
- Facuce





Sales example

- There are stores that put product on sales for Christmas

	store	weeks_to_xmas	avg_week_sales	is_on_sale	weekly_amount_sold
0	1	3	12.98	1	219.60
1	1	2	12.98	1	184.70
2	1	1	12.98	1	145.75
3	1	0	12.98	0	102.45
4	2	3	19.92	0	103.22
5	2	2	19.92	0	53.73

where

  - `store`: id of the store
  - `weeks_to_xmas`: how many weeks before Christmas the measurement was done
  - `avg_week_sales`: how much they sold in average
  - `is_on_sale`: if store puts product on sale
  - `weekly_amount_on_sale`: how much that store has sold in that week

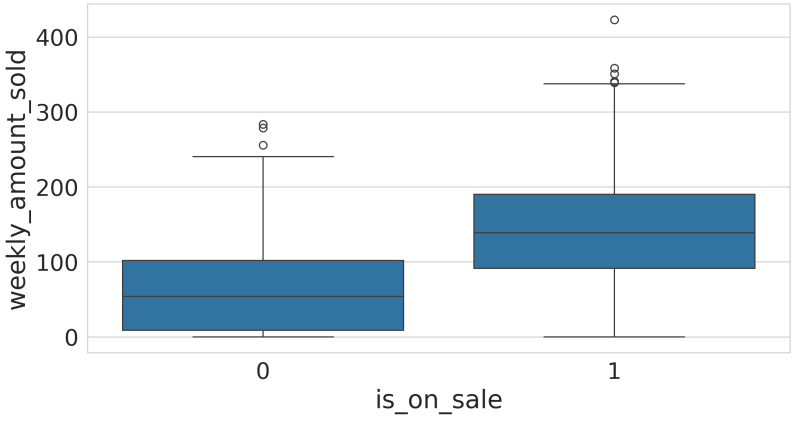
- Causal inference question: *"what is the impact of T on Y?"*
  - E.g., what is the impact of `is_on_sale` on `weekly_amount_sold`?
  - E.g., when is the best moment to have sales (i.e., `weeks_to_xmas`)?

Definition

- **Treatment for unit i**
  - $T_i = 1$ if unit i has received the treatment
  - E.g., `weeks_to_xmas` `is_on_sale`
  - Sometimes T is denoted with D
- **Outcome**
  - The variable one wants to influence, Y_i
  - E.g., `weekly_amount_sold`
- **Unit of analysis**
  - The thing you wish to intervene on / treat
  - E.g., *"people who buy Y"*

The Fundamental Problem of Causal Inference

- **The Fundamental Problem of Causal Inference**
  - *"You can never observe the same unit with and without treatment"*
- **Example**
- Plot outcome `weekly_amount_sold` by treatment `is_on_sale`
- Stores with price drops sell more (from 50 to 150)
- Matches intuition: *"people buy more when prices are low due to sales"*



- Is it true?
  - Maybe only larger businesses afford aggressive price drops?
  - Maybe customers buy close to Christmas anyway?
- Observing both counterfactual situations would reveal the effect of price drops $\rightarrow$ impossible

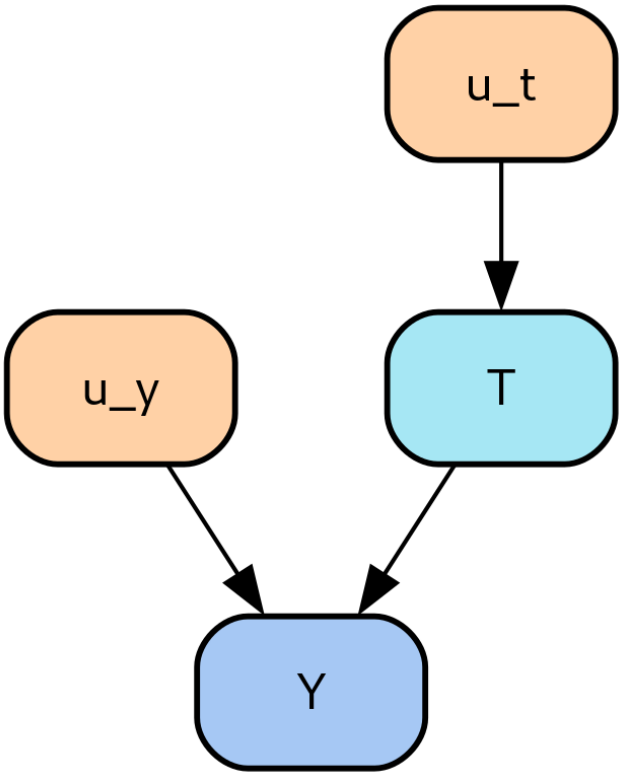
Causal Model

- A **causal model** is a series of assignments $\leftarrow$
  - Causality is non-reversible: you can't rearrange the equation
- **Example**

$$T \leftarrow f_t(u_t)$$
$$Y \leftarrow f_y(T, u_y)$$

where:

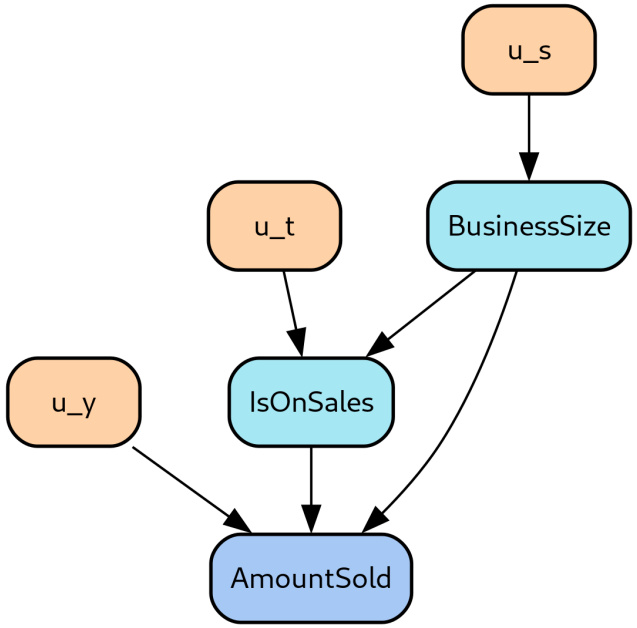
- u_t, u_y are *exogenous* variables (outside the model)
  - Account for everything not covered by model variables
- Treatment T and u_y cause outcome Y
- T and Y are *endogenous* variables (inside the model)



Causal Model: Sales Example

- Causal model for sales example adding a *BusinessSize*

$$BusinessSize \leftarrow f_s(u_s)$$
$$IsOnSales \leftarrow f_t(BusinessSize, u_t)$$
$$AmountSold \leftarrow f_y(IsOnSales, Business$$



- One can make an assumption of linear model for $f_y(.)$ and say

$$AmountSold = \alpha + \beta_1 IsOnSales + \beta_2 BusinessSize + \varepsilon$$

Applying a Treatment to Sales Example

- Causal model is

$$\begin{cases} T \leftarrow f_t(u_t) \\ Y \leftarrow f_y(T, u_y) \end{cases}$$

- Ask *"what would happen to the outcome Y if I were to set the treatment to t_0?"*

$$\begin{cases} T \leftarrow t_0 \\ Y \leftarrow f_y(T, u_y) \end{cases}$$

  - The intervention is represented by the $do()$ operator, e.g., $do(T = t_0)$
- Note that:

$$\mathbb{E}[AmountSold|isOnSales = 1] \neq \mathbb{E}[AmountSold|do(IsOnSales = 1)$$

  - The **first** is what happens to businesses that make a price cut (e.g., a big business)
    - Some business (e.g., big businesses) are "self-selecting" to cut prices
  - The **second** is what happens if you force every business to engage in sales

Individual Treatment Effect

- Express the impact of the treatment T on the outcome Y for an individual unit i

$$\tau_i = Y_i|do(T = t_1) - Y_i|do(T = t_0)$$

  - The effect τ_i of going from treatment t_0 to t_1 for unit i is the difference in the outcome of that unit under t_1 compared to t_0
  - E.g., $AmountSold_i|do(IsOnSales = 1) - AmountSold_i|do(IsOnSales = 0)$
- You can only observe one term due to the fundamental problem of causal inference
  - Represent it in theory, but can't recover it from data

Potential Outcomes

- Aka "counterfactuals"
- The $do()$ operator represents *"unit i outcome is Y if treatment is set to t"*

$$Y_{t,i} = Y(t)_i = Y_i | do(T_i = t)$$

  - The first is "Rubin's notation", the second is "Pearl's notation"
- For binary treatment use $Y_{0,i}$ (without treatment) vs $Y_{1,i}$ (with treatment)
  - One is "factual" (i.e., observable)
  - The other is "counterfactual" (i.e., theoretical, not observable)
- There is a slight difference
  - $Y_t = Y | do(T = t)$ is "what a business would sell if it cut prices"
  - $Y | T = t$ is "what a business that cut the prices sold"

Consistency of the Treatment

- There is a single version of the treatment
  - I.e., $T = T_i$ means only one thing
- E.g.,
  - A drug can come in multiple dosages: only one is used
  - Giving a discount coupon is treated as the value of the coupon is the same, it's given only once, in a way that is always equivalent
  - In "help from a financial planner", is "help" a one-time consultation or regular advice?

No Interference

- Aka "Stable Unit of Treatment Value (SUTVA)"
- The effect of one unit is not influenced by the treatment of other units
  - E.g., there are no spillovers or network effects
- E.g., vaccinating one person makes other close people less likely to catch the illness even if they don't get the treatment

Average Treatment Effects

- Assume you can see both potential outcomes Y_{1t} and Y_{0t} at the same time (which is impossible)
- **Average Treatment Effect (ATE)** is the impact that the treatment T would have on the average:

$$ATE = \mathbb{E}[Y_{1t} - Y_{0t}]$$

  - You can define it, but not measure it
  - E.g., average impact of price cuts on amount is:

$$ATE = \mathbb{E}[AmountSold_{1i} - AmountSold_{0i}]$$

- **Average Treatment effect on the Treated (ATT)** is the impact of the treatment on the units that got the treatment:

$$ATT = \mathbb{E}[Y_{1i} - Y_{0i} | T = 1]$$

  - You can define it, but not measure it
  - E.g., how much sales increase for the business engaged in price cuts:

$$ATT = \mathbb{E}[AmountSold_{1i} - AmountSold_{0i} | IsOnSales = 1]$$

- **Conditional Average Treatment Effects (CATE)** is the impact of the treatment on the units in a group defined by $X = x$

Bias

- Bias is what makes association different from causation
- The claim *"cutting prices increases the amount sold by a business"* can be questioned with
  - *"Those businesses that cut prices would probably have sold more anyway, even with no price cuts"*
- More formally the argument refers to potential outcomes
  - We are trying to estimate $\mathbb{E}[Y_t] = \mathbb{E}[Y|do(T = t)]$ using $\mathbb{E}[Y|T = t]$ but this is a "biased estimator" (i.e., an incorrect one)

The Bias Equation

- Causal inference is interested in:

$$\mathbb{E}[Y_1 - Y_0] = \mathbb{E}[Y|do(T=1)] - \mathbb{E}[Y|do(T=0)]$$

- What is measured is:

$$\mathbb{E}[Y|T=1] - \mathbb{E}[Y|T=0]$$

- Let's express one in terms of the other
  - Replacing the observed outcomes (each unit has two potential outcomes but only one can be observed and the other is 0)

$$Y|T=1 = (Y_1|T=1) + (Y_0|T=1) = Y_1|T=1$$

  - Adding and subtracting $\mathbb{E}[Y_0|T=1]$, reordering and merging expectations

$$\mathbb{E}[Y|T=1] - E[Y|T=0] = \mathbb{E}[Y_1 - Y_0|T=1] + \underbrace{(\mathbb{E}[Y_0|T=1] - \mathbb{E}[Y_0|T=0])}_{\text{Bias}}$$

Association = causation + a bias term

The Bias Equation: Comments

- Association = causation + a bias term

$$\mathbb{E}[Y|T=1]-E[Y|T=0] = \mathbb{E}[Y_1-Y_0|T=1]+\underbrace{(\mathbb{E}[Y_0|T=1]-\mathbb{E}[Y_0|T=0])}_{\text{Bias}}$$

- The bias is given by how the treated and control group differ regardless of the treatment
  - Many things you can't observe are changing together with the treatment
  - The treated and untreated businesses differ in more ways than just cutting prices
- Remark 3:
  - For association to be causation, treated and control units would have to be exchangeable

The bias equation: visual example

- Plot the outcome `weekly_amount_sold` by business size measured by `avg_week_sales`
- Color the points by treatment (red) / control (blue)
- The treated units are more concentrated to the right of the plot